

Jialing Dai

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Assistant Professor, Central University of Finance and Economics
Ph.D., UCAS | Former Visiting Researcher, The University of Texas at Dallas
Algorithm design and analysis; combinatorial optimization and approximation algorithms; network science and computational social systems; graph learning and reinforcement learning; data-driven decision-making

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- **National recognition.** Selected for the **inaugural 2024 doctoral sponsorship cohort** of the China Association for Science and Technology (CAST). Recipient of the **National Scholarship** for graduate students and **two National Scholarships** for undergraduate students.
- **Publications. 10** peer-reviewed papers published or accepted, including **5 first-author journal articles** and **1 corresponding-author conference paper**. First-author work appears in IEEE Transactions on Networking, Information Sciences, Expert Systems with Applications, and Chinese Journal of Management Science.
- **Research independence.** Led studies from **problem formulation and mathematical modeling** through **complexity and approximation analysis**, **algorithm design**, implementation, **computational experiments**, visualization, and manuscript preparation.
- **International collaboration.** Conducted visiting research at The University of Texas at Dallas with Weili Wu and Ding-Zhu Du. Led the proposal and delivery of a funded International Young Partners project connecting researchers at UCAS and the University of Massachusetts Boston.

EDUCATION AND RESEARCH APPOINTMENTS

Sep. 2020 – Oct. 2026 — **University of Chinese Academy of Sciences (UCAS)** — Beijing, China

Ph.D. in Computer Applied Technology. Began the integrated graduate program in September 2020 and entered the doctoral stage in March 2023.
Advisor: Prof. Jianming Zhu. **GPA: 3.83/4.00**.
Dissertation: Opinion Propagation Models and Group-Aware Influence Optimization Algorithms in Social Networks.
Graduated and received the Ph.D. on October 16, 2026.

Jan. 19, 2025 – Jan. 26, 2026 — **The University of Texas at Dallas** — Richardson, TX, USA

Visiting Researcher, Department of Computer Science.
Faculty hosts: Profs. Weili Wu and Ding-Zhu Du. Collaborative research on graph algorithms, robust coverage, resource diffusion, and networked decision models.

Sep. 2016 – Jun. 2020 — **Beijing University of Chemical Technology (BUCT)** — Beijing, China

Bachelor of Management, Information Management and Information Systems.
Average: **92.68/100**; **ranked first in the major**. Training in information systems, data analytics, operations research, and management.

SELECTED FIRST-AUTHOR PAPERS

[1] **Jialing Dai**, Yisheng Zhou, Yefeng Sun, Jianming Zhu, Weili Wu.
"Unified Group-Aware Influence Maximization with Generalized Deep Reinforcement Learning."
IEEE Transactions on Networking (**CCF A, JCR Q1**), 34: 6198–6213, 2026. [DOI ↗]

Developed UGIP, a **unified, parameterized framework for group-aware influence propagation** that represents echo-chamber, topic-aware, homophily, and higher-order effects within a common modeling interface. Established **NP-hardness and #P-hardness** results and analyzed when group interactions invalidate submodularity and supermodularity. Designed UG-DCGNN, which combines coupled **graph neural networks with deep reinforcement learning** (Double DQN) and incorporates group evidence through input-side gated fusion to learn sequential seed-selection policies. The architecture is reusable

across propagation settings, with a separate policy trained for each setting; **inference requires no Monte Carlo simulation**. Evaluated influence quality and computational efficiency on synthetic and real networks, with ablation and sensitivity studies examining group inputs, sequential re-scoring, marginal-gain rewards, signed descriptors, and alignment between group structure and network topology.

[2] **Jialing Dai**, Jianming Zhu, Guoqing Wang.
"Competitive net influence maximization on intergroup debate effect."
Information Sciences (**CAS Zone 1 (Top), JCR Q1**), 680: 121139, 2024. [DOI ↗]

Developed the Intergroup Debate Independent Cascade (IDIC) model to explain how **feedback between group consensus and individual behavior** can amplify both discussion and negative influence. Used a **group hypergraph** to couple cross-group interactions with member-level propagation, and formulated the Controversial Net Influence Maximization (CNIM) problem. Established **NP-hardness and non-submodularity**, constructed submodular upper and lower bounds, and applied a sandwich framework to obtain **data-dependent approximation guarantees**. Introduced a heterogeneous competitive cascade model with information-type-dependent activation probabilities, together with **reverse-sampling-based estimation** and the HCISG seed-selection algorithm. Improved efficiency through sampling refinement, tailored data structures, and pruning. Experiments on synthetic and real networks assessed solution quality and scalability and showed how stronger group cohesion can amplify debate-driven information diffusion.

[3] **Jialing Dai**, Jianming Zhu, Guoqing Wang.
"Opinion influence maximization problem in online social networks based on group polarization effect."
Information Sciences (**CAS Zone 1 (Top), JCR Q1**), 609: 195–214, 2022. [DOI ↗]

Built a **Zhihu dataset of 50 discussion threads and 127,178 replies** to investigate group polarization and inform the modeling of online opinion propagation. Proposed the Enhanced Independent Cascade (EIC) model, combining **individual preference updates**, positive and negative activation, and group-level polarization. The model captures how exposure to persuasive information can reinforce agreement or provoke an opposing response when preferences differ substantially. Developed GEIM, a **group-based seed-selection algorithm** that combines structure-aware candidate screening, local influence estimation, and iterative seed adjustment. Analyzed computational and structural properties and evaluated how initial preferences, group size, and the surrounding opinion environment affect diffusion outcomes. The experiments show that **broader propagation can be counterproductive in strongly polarized groups**, reducing rather than improving net positive influence.

[4] **Jialing Dai**, Peikun Ni, Ke Su, Jianming Zhu, Weili Wu, Ding-Zhu Du.
"Influence Maximization in Online Social Networks: Models, Algorithms, and Trustworthy Decision-Making."
Expert Systems with Applications (**CAS Zone 1 (Top), JCR Q1, IF 9.4**), article 134131, 2026. Review article.
[DOI ↗]

Co-developed the **survey framework and taxonomy**, synthesized the literature, and drafted and revised the manuscript. Organized influence maximization around the relationships among network representations, behavioral assumptions, diffusion models, optimization objectives, algorithmic methods, and decision requirements. Compared greedy and lazy-greedy algorithms, sampling and sketching, mathematical programming, non-submodular search, learning-based seed selection, and adaptive policies. Examined how group effects, competition, changing networks, and objectives beyond node coverage alter the applicability of existing methods. Framed **trustworthy influence decision-making** in terms of robustness, fairness, privacy, harm control, causality, accountability, and reproducible evaluation. The resulting synthesis clarifies modeling assumptions, **theoretical guarantees, evaluation evidence, and application boundaries**, providing a methodological framework for research on network and platform decision systems.

[5] **Jialing Dai**, Yefeng Sun, Ding-Zhu Du, Weili Wu, Jianming Zhu.
"Orthant-Submodular Maximization: Polynomial-Time Approximation and Sharp Asymptotics."
Operations Research **(UTD24, FT50, AJG 4*)**, 2026. **Under Review.**

Studied discrete allocation decisions in which each resource receives one label or remains unused and additional actions can reduce total value, modeled by nonnegative orthant-submodular functions without cross-label monotonicity. Developed dynamic-overlap search, a **uniform polynomial-time randomized**

algorithm that combines candidate-dependent comparison paths with joint gain–return estimates to derive separation inequalities valid at an unknown optimum, using only value queries at feasible assignments. Established **first-order matching between algorithmic guarantees and information-theoretic bounds**. For both polynomial-time and polynomial-query algorithms, derived the optimal-ratio expansion $\beta_{\mathrm{G}} + c_{+}/k + O(k^{-2})$, where k is the number of labels, $\beta_{\mathrm{G}} \approx 0.15749$ and $c_{+} \approx 0.11295$. The information upper bound holds even when the set of resources used by an optimum is given in advance. Also characterized the **exact two-label ratio of 1/4** and developed a support-sensitive three-label algorithm attaining $V_{\mathrm{supp}} - \varepsilon$, where $V_{\mathrm{supp}} \approx 0.18225$, using **at most 3n value queries** for n resources and any prescribed accuracy loss $\varepsilon > 0$.

[6] **Jialing Dai**, Yefeng Sun, Ding-Zhu Du, Ke Su, Weili Wu, Jianming Zhu.
"Refined Label Comparisons for Constrained Non-Monotone k-Submodular Maximization."
International Conference on Learning Representations (ICLR, **CCF A, ICORE A***), 2027.* **Under Review.**
[OpenReview ↗]

Develops **randomized greedy and continuous algorithms** for constrained non-monotone k-submodular maximization and, more generally, full-domain orthant-submodular objectives with r-wise monotonicity. Introduces Random-root, a locally minimax randomized greedy method with a **closed-form approximation guarantee under p-system constraints**. Under matroid constraints, exploits label-specific replacement losses to improve the k-submodular approximation coefficient from $\sqrt{2}-1\,(\approx 0.4142)$ to $\alpha_2 - \tau$, where $\alpha_2 \approx 0.4218$ and $\tau > 0$ is an arbitrarily small accuracy loss. Characterizes how additional labels strengthen the guarantees, including $0.4366 - \tau$ for three to five labels and a uniform $0.4259 - \tau$ for all $k \geq 3$. Establishes finite randomized implementations, with $O(nk\rho)$ **value queries** for the k-submodular baseline at fixed accuracy, where n is the ground-set size and ρ the matroid rank. Extends the guarantees to real-valued objectives measured relative to the empty assignment.

[7] **Jialing Dai**, Jianming Zhu, Guoqing Wang, Jun Huang.
"Net Positive Information Diffusion Activity Maximizing in Signed Online Social Networks."
Chinese Journal of Management Science (**CSSCI, CSCD, PKU Core Journals**) 33(3): 139–150, 2025. [In Chinese.] [DOI ↗]

Formulated net-positive information activity maximization under a Signed Independent Cascade model, **jointly selecting seed nodes and their initial positive or negative states**. Measured diffusion benefits through edge-level positive and negative interactions, rather than the number of reached users alone. Established NP-hardness and #P-hardness and analyzed the interaction effects that make the original objective non-monotone and neither submodular nor supermodular. Constructed a **path-based submodular coverage surrogate** using signed propagation paths and approximate path gains, then developed a **maximum-coverage greedy algorithm** with CELF acceleration. Experiments on real signed networks distinguish reach from beneficial activity and show why a strategy that **reaches more users need not produce better net-positive word-of-mouth outcomes**.

OTHER JOURNAL PAPERS

[8] Yefeng Sun, **Jialing Dai**, Liang Gong, et al. "CaMT: Class-aware Multi-level Training Distillation for Lightweight Semantic Segmentation in Apple Orchard." *Artificial Intelligence in Agriculture* (**CAS Zone 1 (Top), JCR Q1, IF 16.1**), 2026. [DOI ↗]

[9] Yefeng Sun, **Jialing Dai**, Liang Gong, et al. "PROMO-Orchard: A Robust Semantic-Enhanced Framework for Dual-Robot Simultaneous Localization and Mapping Merging in Orchard." *Journal of Field Robotics* (**CAA A**) 43(7): 4880–4903, 2026. [DOI ↗]
Featured on the cover of the October 2026 issue.

[10] Yefeng Sun, Wei Zhang, **Jialing Dai**, et al. "A Robot-Edge-Cloud Orchestration Framework for Heterogeneous Multi-Robot Systems in Greenhouse Production." *IEEE Robotics and Automation Letters* (RA-L), 2026. **Under Review**.

CONFERENCE PAPERS

[11] Ke Su, **Jialing Dai (corresponding author)**, Weili Wu. "Scenario-Robust Checkpoint Cover for AI-Generated Content Provenance." *International Conference on Algorithmic Aspects in Information and Management (AAIM)*, 2026. Accepted / Springer LNCS (**EI Compendex** series).

Contributed to **robust checkpoint-placement modeling** for AI-generated content provenance under uncertain source and propagation scenarios, and to a **reverse-sampling-based solution** with bicriteria and high-probability guarantees.

[12] Yisheng Zhou, **Jialing Dai**, Dongyu Mao, Ke Su, Weili Wu. "Modeling Organizational Performance through Competitive Resource Diffusion." *International Conference on Algorithmic Aspects in Information and Management (AAIM)*, 2026. Accepted / Springer LNCS (**EI Compendex** series).

[13] Yefeng Sun, Liang Gong, **Jialing Dai**, Bishu Gao, et al. "S²BEV: Lightweight, Robust, and Precise SLAM-Oriented Segmentation Bird Eye's View Mapping Approach." *IEEE International Conference on Robotics and Automation (ICRA,* **CCF B, ICORE A***), Atlanta, Georgia, USA, 2025, pp. 13160–13165. [DOI ↗]

RESEARCH FUNDING AND PROJECTS

CAST Young Elite Scientists Sponsorship Program — Doctoral Student Special Program Jan. 2025 – Dec. 2026

Awardee, inaugural 2024 cohort

China Association for Science and Technology (CAST)

Selected for a national doctoral sponsorship providing academic funding and research-development support. The award supports participation in academic conferences, publication, and international research exchange, together with academic mentoring and professional development. Doctoral research focuses on opinion propagation and group-aware influence optimization, linking mathematical modeling, algorithmic analysis, and computational evaluation.

2025 International Young Partners Program Dec. 2025 – May 2026

Proposal and Project Lead | Completed

Foundation for the Development of Science and Technology in China (FDSTC)

Led proposal development and delivery of a funded international academic exchange, including a five-day visit to UCAS by Dr. Hongmin (William) Du of the University of Massachusetts Boston, March 16–20, 2026. Designed the academic agenda, coordinated seminars and research discussions, and completed project reporting and closeout. The exchange covered large language models, network propagation, cybersecurity information disclosure, and data-driven risk analytics, with discussions of joint research questions, modeling approaches, and experimental design.

Risk and Safety Analytics for Power-Grid Construction Oct. 2022 – Dec. 2022

Student Team Lead, under faculty supervision

State Grid Henan Electric Power Research Institute

Led a six-member student team in an applied analytics collaboration. Developed a hierarchical safety taxonomy and a 1,498-case labeled dataset from accident records and safety-management materials. Conducted association-rule and clustering analyses, interpreted recurring risk patterns, and drafted targeted management recommendations for construction-safety scenarios.

Hazard-Source Tracing with Fault-Tree Modeling May 2022 – Apr. 2023

Student Team Co-Lead, under faculty supervision

Developed a hazard-source tracing approach based on fault-tree analysis, organized causal structures from heterogeneous incident evidence, and completed multiple case studies. Wrote major sections of the final report, connecting formal risk models with case-based analysis and management recommendations.

TEACHING AND MENTORING EXPERIENCE

Teaching Assistant 2021–2023
University of Chinese Academy of Sciences, School of Engineering Science
Mathematical Methods and Their Applications in Management: 127 students; course team of one instructor and two teaching assistants. Accounting and Cost Management: 74 students; one instructor and one teaching assistant. Human Factors Engineering: 14 students; one instructor and one teaching assistant. Supported tutorials, case discussions, student projects, and office hours. Assisted with assignments, assessment rubrics, homework and examination grading, course logistics, and technical support.

Peer Instructor 2018–2020
Beijing University of Chemical Technology, Peer Academic Development Center
Provided course-related instruction and peer academic support for undergraduate students. Explained technical material, guided problem-solving activities, and organized peer-learning sessions.

- **Book and teaching-material support** | 2023 – 2024. Under Prof. Jianming Zhu's supervision, assisted with drafting selected sections of Chapter 7, preparing figures, and formatting manuscript materials for his monograph *Model, Algorithm, Application of Information Dissemination in Online Social Networks* (China Machine Press, 2025; in Chinese); translating portions of Chapters 9–10 ("Integer Linear Programming" and "Heuristic and Constraint Programming") and LaTeX typesetting for *Operations Research: An Introduction*, 11th edition (Chinese translation not yet published); and expanding selected sections, consolidating manuscript material, and organizing references for *Quantitative Analysis: Models and Methods* teaching materials.

ACADEMIC SERVICE AND LEADERSHIP

- **Program Committee Member**: 20th International Conference on Algorithmic Aspects in Information and Management (AAIM) | 2026
- **Peer Reviewer** (2023 – Present): IEEE Transactions on Networking; Expert Systems with Applications; IEEE Transactions on Computational Social Systems; World Wide Web.
- **Research Seminar and Journal-Club Coordination** (UCAS | 2020 – 2025): Presented and discussed research in weekly seminars, later taking responsibility for paper selection, pre-presentation review, and meeting moderation. Prepared agendas, documented discussions, and provided methodological feedback to master's and doctoral researchers.
- **Graduate Program and Admissions Support** (UCAS | 2022 – 2023): Coordinated master's thesis defenses and graduate admissions interviews, including scheduling, documentation, records management, and communication among faculty, students, and administrative staff.
- **Student Leadership and Campus Service** (UCAS and BUCT | 2017 – 2023): Served as graduate student union president and class president at UCAS, and as class president and student-organization officer at BUCT. Responsibilities included student representation, activity coordination, and program support. Also worked as a library assistant at the UCAS Yuquan Road Library.

HONORS AND AWARDS

CAST Young Elite Scientists Sponsorship Program — Doctoral Student Special Program 2024
National-level doctoral research sponsorship · Inaugural cohort · China Association for Science and Technology

National Scholarship for Graduate Students 2022
National-level academic merit scholarship · Ministry of Education, China

National Scholarship for Undergraduate Students 2018, 2019
National-level academic merit scholarship · Ministry of Education, China · **Awarded twice**

Outstanding Young Scholar Paper Award, Second Prize 2021
International conference paper award · International Symposium on Emergency Management

Outstanding Graduate of Beijing 2020
Beijing Municipal Education Commission.

Last updated: Nov. 9, 2026

Merit Student 2017, 2022, 2023
BUCT, 2017; UCAS, 2022 and 2023.

Excellent Student Leader 2018, 2022
BUCT, 2018; UCAS, 2022.

COURSES PREPARED TO TEACH

- **Algorithms and mathematical foundations**: Programming Fundamentals; Algorithms and Data Structures; Discrete Mathematics; Probability and Statistics; Computational Complexity.
- **Optimization and quantitative decision methods**: Operations Research; Optimization Models and Methods; Combinatorial Optimization; Mathematical Modeling; Decision Analytics and Management Decision Modeling.
- **Data science and information systems**: Python Programming and Data Analysis; Data Mining; Machine Learning; Database Systems; Data Visualization; Management Information Systems; Data Governance.
- **Networks and intelligent decision systems**: Social and Complex Network Analysis; Trustworthy AI and Intelligent Decision-Making.

RESEARCH METHODS, TECHNICAL SKILLS, AND LANGUAGES

- **Research methods**: Problem formulation and mathematical modeling; complexity and approximation analysis; combinatorial optimization; submodular, non-submodular, and non-monotone optimization; graph algorithms; sampling and greedy algorithms; robust coverage and resource allocation; graph neural networks; deep reinforcement learning.
- **Computational research**: Data cleaning and analysis; Monte Carlo simulation; algorithm implementation; experimental design; ablation, sensitivity, and statistical analysis; visualization; reproducible computational workflows; technical and academic writing.
- **Programming**: Python (PyTorch, NetworkX, NumPy, pandas), MATLAB, C, Java, and SQL.
- **Data and research tools**: MySQL, SQL Server, MongoDB, Hadoop, SPSS, LaTeX, Git, Linux, and Jupyter.
- **Languages**: Mandarin Chinese (native); English (professional working proficiency).

EARLIER RESEARCH AND TECHNICAL PROJECTS

- **Multi-objective flexible job-shop scheduling** (Undergraduate work, BUCT | 2017 – 2020): Implemented an improved NSGA-II with niche preservation, elitism, adaptive operators, and greedy initialization. Evaluated scheduling solutions using Gantt charts and Pareto-front analysis.
- **Rail-guided vehicle scheduling** (Undergraduate work, BUCT | 2017 – 2020): Modeled periodic RGV scheduling with an improved genetic algorithm that jointly considered tool allocation, vehicle sequencing, and batch-size decisions. Compared performance with a standard genetic algorithm baseline.
- **Mathematical modeling competitions** (Undergraduate work, BUCT | 2017 – 2020): Developed cellular-automaton models of SIS-style opioid spread and support-vector-machine models for traffic-congestion prediction.
- **Hadoop Data Wall** (Undergraduate work, BUCT | 2018 – 2020): Served as student project lead for a service-outsourcing competition entry. Configured a pseudo-distributed Hadoop environment, built an extract–transform–load pipeline into MySQL, and delivered an interactive dashboard demonstration.
- **Advisor-selection web system** (Undergraduate work, BUCT | 2018 – 2020): Built a Java JSP/Servlet and MySQL application supporting role-based access, preference submission, a matching workflow, form validation, database persistence, and create/read/update/delete operations.
- **Financial-distress prediction** (Undergraduate work, BUCT | 2018 – 2020): Completed a course research project combining Information Gain and ReliefF feature selection with self-organizing-map pre-selection and a backpropagation neural-network classifier.
- **Artificial Intelligence and Future Technologies Programme** (National University of Singapore, School of Continuing and Lifelong Education | Feb. 11–15, 2019): Completed a short-term study program and received a certificate of completion. Project work included setting up and evaluating a classification pipeline using Azure Face API.